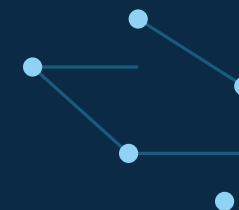




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ROYAL UNIVERSITY OF PHNOM PENH



FACULTY OF ENGINEERING

Department of Information Technology Engineering

Project-Practicum NE

Lecturer: So Pheaktra

ប្រធានបទបរិយាកាសបណ្តាញស្របច្បាប់

High-Availability Campus Network Design &

Implementation: Secure Layer 3 Redundancy using

HSRP and Inter-VLAN Routing

Prepared by

Seng Heng

Tong Borey



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1. Introduction

This project presents the design and simulation of a hierarchical campus network. By combining VLAN segmentation, First Hop Redundancy (HSRP), and automated address management (DHCP), the network is built to stay available even when a core device fails, while keeping traffic between departments properly isolated and secured.

Core technologies combined in this project

- VLAN segmentation — isolates Student, Staff, and Management traffic
- HSRP — first-hop gateway redundancy between two core switches
- DHCP — automated, split-scope IP address management



Hierarchical Design

Core, distribution, and access layers separate roles for clarity and control.



Layer-3 Redundancy

Two core switches share gateway duty so no single device is a point of failure.



Built-in Security

Port security and access control reduce exposure on an open campus network.



2. Problem Statement

Despite the availability of advanced networking protocols, many educational institutions still rely on flat, fragile network designs. The following technical deficiencies are commonly observed in legacy campus networks:

1

Single Point of Failure

One core switch outage takes the entire network offline.

2

Inefficient IP Management

Manual, static addressing is slow and error-prone at scale.

3

Security Vulnerabilities

Open access ports allow unauthorized devices onto the network.

4

Lack of Hierarchy

Flat designs don't scale and make troubleshooting difficult.



3. Project Objectives



Ensure continuous network availability during device failure



Improve network performance and efficiency



Build a reliable and resilient campus infrastructure



Enable a scalable, future-ready network design



Enhance network security in an open campus environment



Provide a cost-effective, practical blueprint for universities



4. Project Scope

1

Campus LAN Focus

2

Hierarchical Network Architecture

3

Layer 2 Network Technologies

4

Layer 3 Network Technologies

5

High Availability & Redundancy

6

IP Addressing & DHCP Management

7

Network Security Measures

8

Simulation-Based Implementation

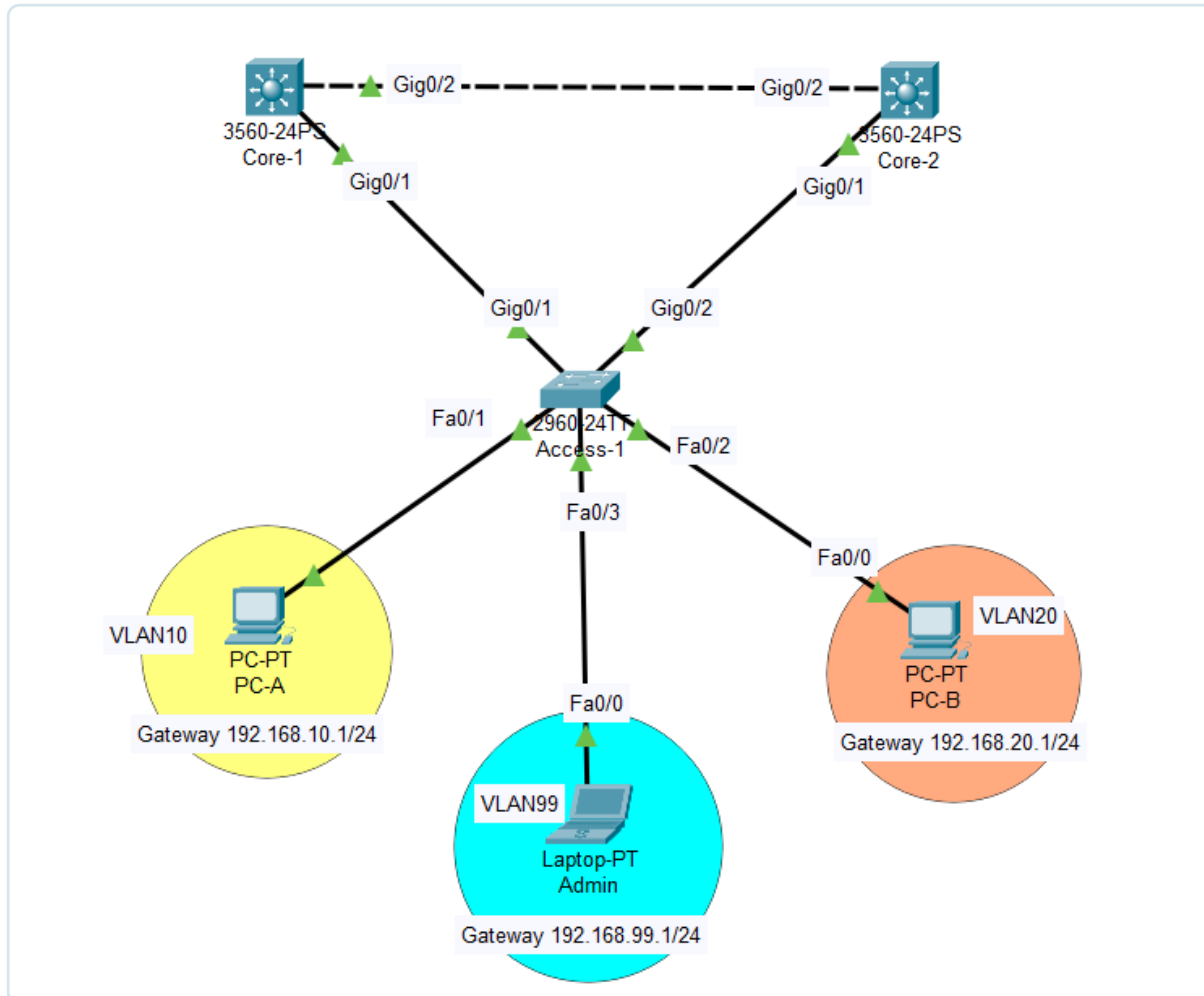
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User Segmentation



5. Network Topology

Two core switches (Core-1, Core-2) provide redundant Layer-3 gateways for three VLANs, distributed through Access-1 to end devices.



Key Components

- Core-1 / Core-2 — redundant Layer-3 core (HSRP pair)
- Access-1 — access-layer switch, dual-uplinked to both cores
- VLAN 10 (Students), VLAN 20 (Staff), VLAN 99 (Management)
- Each VLAN has its own subnet and HSRP virtual gateway



6. Network Design — VLAN & Addressing

VLAN	Name	Subnet	Core-1 IP	Core-2 IP	Virtual IP (HSRP)	Access Port
10	Students	192.168.10.0/24	192.168.10.2	192.168.10.3	192.168.10.1	Fa0/1
20	Staff	192.168.20.0/24	192.168.20.2	192.168.20.3	192.168.20.1	Fa0/2
99	Management	192.168.99.0/24	192.168.99.2	192.168.99.3	192.168.99.1	Fa0/3

VLAN Membership — Access-1 (show vlan brief)

```
Access-1#show vlan bri
VLAN Name                Status  Ports
-----
1    default              active  Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                   Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                   Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                   Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                   Fa0/21, Fa0/22, Fa0/23, Fa0/24
10   STUDENTS             active  Fa0/1, Fa0/4
20   STAFF                active  Fa0/2
99   MANAGEMENT           active  Fa0/3
1002 fddi-default        active
1003 token-ring-default  active
1004 fddinet-default     active
1005 trnet-default       active
Access-1#
```

Reading the evidence

- Fa0/1, Fa0/2, Fa0/3 map cleanly to VLAN 10, 20, 99
- Each VLAN's gateway is virtual (HSRP), shared by both cores
- Split-scope DHCP on Core-1 and Core-2 keeps addressing redundant too



6. Network Design — Trunking & Spanning Tree

show interfaces trunk — verified on all three switches

Core-1

```
Core-1#show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1
Gig0/2    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,99
Gig0/2    1,10,20,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    10,20,99
Gig0/2    1,10,20,99
```

Core-2

```
Core-2#show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1
Gig0/2    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,99
Gig0/2    1,10,20,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,99
Gig0/2    1,10,20,99
```

Access-1

```
Access-1#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1
Gig0/2    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,99
Gig0/2    1,10,20,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,99
Gig0/2    1
```



Why redundancy is safe here: Spanning Tree Protocol

The topology forms a physical loop (Core-1 ↔ Core-2 ↔ Access-1 ↔ Core-1). PVST+ automatically blocks the redundant path per-VLAN to prevent a loop — e.g. Access-1's link to Core-2 forwards only VLAN 1 under normal conditions — while keeping that link ready to take over instantly if the primary path fails.



7. HSRP — First Hop Redundancy

Core-1 and Core-2 share a single virtual gateway IP per VLAN. Core-1 is configured with priority 150 and preempt, so it always becomes Active when available; Core-2 holds priority 100 (default) as Standby.

CORE-1 — ACTIVE

```
Core-1#show standby bri
P indicates configured to preempt.
|
Interface  Grp  Pri P State   Active        Standby        Virtual IP
Vl10       1    150 P Active   local         192.168.10.3   192.168.10.1
Vl20       20   150 P Active   local         192.168.20.3   192.168.20.1
Vl99       99   150 P Active   local         192.168.99.3   192.168.99.1
```

CORE-2 — STANDBY

```
Core-2#show standby brief
P indicates configured to preempt.
|
Interface  Grp  Pri P State   Active        Standby        Virtual IP
Vl10       1    100 Standby  192.168.10.2   local         192.168.10.1
Vl20       20   100 Standby  192.168.20.2   local         192.168.20.1
Vl99       99   100 Standby  192.168.99.2   local         192.168.99.1
```

150

Core-1 Priority

100

Core-2 Priority

Preempt: ON

Core-1 reclaims Active automatically



8. Security

- Port security with sticky MAC on all access ports (Fa0/1–0/3)
- All unused switch ports administratively shut down
- Remote management restricted to SSH only (telnet disabled)
- Passwords and secrets encrypted (service password-encryption)
- Login banner warns against unauthorized access

```
Access-1 - interface configuration

interface FastEthernet0/1
  switchport access vlan 10
  switchport port-security
  switchport port-security mac-address sticky
!
interface FastEthernet0/3
  switchport access vlan 99
  switchport port-security
  switchport port-security mac-address sticky
!
line vty 0 4
  transport input ssh
```



All three access ports — Student, Staff, and Management — are secured consistently. The Management port carries the most sensitive traffic, so it follows the same sticky-MAC port-security policy as the others rather than being left open.



9. Testing & Results — Normal Operation

BASELINE PING TEST

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

CONTINUOUS PING STARTED

```
C:\>ping -t 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
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Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
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Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
```

Student PC → Staff Gateway (192.168.20.1): 4/4 replies, 0% loss

Test Setup

- Confirmed normal connectivity across VLANs before introducing failure
- Core-2 confirmed Standby for all VLANs
- Continuous ping (ping -t) started to a Staff-VLAN host to observe failover live



9. Testing & Results — Failover Event

Core-1 was powered off to simulate a hardware failure.

CORE-2 — NOW ACTIVE

```
Core-2#show standby brief
                P indicates configured to preempt.
                |
Interface Grp Pri P State Active Standby Virtual IP
Vl10      1   100   P Active local unknown 192.168.10.1
Vl20      20  100   P Active local unknown 192.168.20.1
Vl99      99  100   P Active local unknown 192.168.99.1
```

All 3 VLANs flip to Active on Core-2; Standby shows “unknown” — Core-1 is unreachable

LIVE PING — THE FAILOVER GAP

```
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=7ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
```

5 dropped packets (a few seconds) before traffic automatically resumes — no manual intervention needed.



9. Testing & Results — Recovery

Core-1 was powered back on. With preempt enabled, it automatically reclaims the Active role.

CORE-1 — STATE LOG & ACTIVE AGAIN

```
Core-1(config-if)#
%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan10 Grp 1 state Standby -> Active

Core-1(config-if)#
%HSRP-6-STATECHANGE: Vlan99 Grp 99 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan99 Grp 99 state Standby -> Active

Core-1(config-if)#
%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Standby -> Active

Core-1(config-if)#exit
Core-1(config)#exit
Core-1#
%SYS-5-CONFIG_I: Configured from console by console

Core-1#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri P State    Active        Standby        Virtual IP
Vl10         1   100 Standby  192.168.10.2   local          192.168.10.1
Vl20         20  100 Standby  192.168.20.2   local          192.168.20.1
Vl99         99  100 Standby  192.168.99.2   local          192.168.99.1
```

CORE-2 — BACK TO STANDBY

```
Core-2#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri P State    Active        Standby        Virtual IP
Vl10         1   100 Standby  192.168.10.2   local          192.168.10.1
Vl20         20  100 Standby  192.168.20.2   local          192.168.20.1
Vl99         99  100 Standby  192.168.99.2   local          192.168.99.1
```

5 pkts

Dropped during failover window

100%

Automatic recovery, no manual steps



10. Conclusion



This project successfully designed and validated a high-availability campus network combining VLAN segmentation, HSRP gateway redundancy, split-scope DHCP, and port-level security. Live failover testing confirmed automatic recovery within a few seconds of packet loss and zero manual intervention, and preempt correctly restored Core-1 to Active once it returned online. An early test also revealed a real engineering trade-off: classic PVST's slower convergence noticeably extended outage time, motivating a move toward Rapid PVST for faster recovery in future iterations.

Suggested Future Work



Migrate to Rapid PVST (or MST) for sub-second STP convergence



Add dynamic routing (e.g. OSPF) for multi-building scalability



Introduce AAA/RADIUS authentication for administrative access

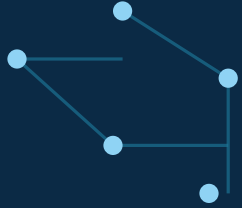


Add SNMP-based monitoring for proactive failure detection



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Thank You

High-Availability Campus Network Design & Implementation

Seng Heng · Tong Borey

